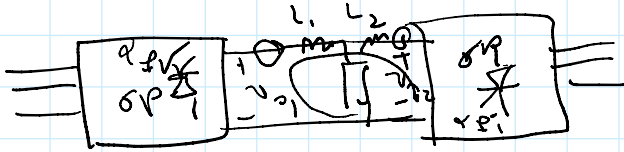
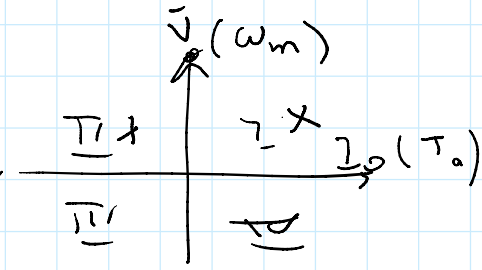
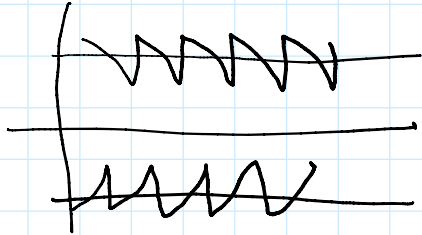


thyristor-based rectifiers

6 pulse - $\bar{V}_o \approx \frac{3}{\pi} V_{rep} \cos \alpha$ ✓

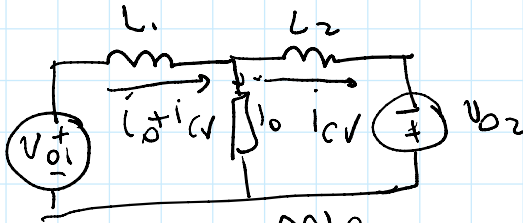
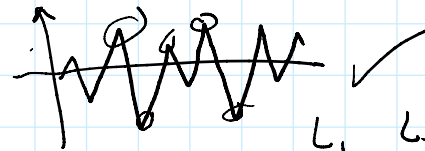


$\bar{V}_o = \bar{V}_{o2} \Rightarrow \alpha_{fV} + \alpha_{f_i} = 180^\circ$



$V_{o1}, \Delta V_o = V_{o1} + V_{o2}$

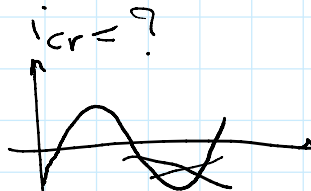
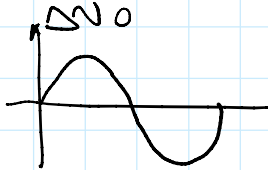
$i_{cr} = ?$



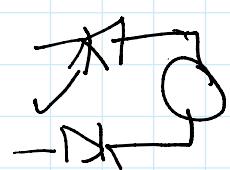
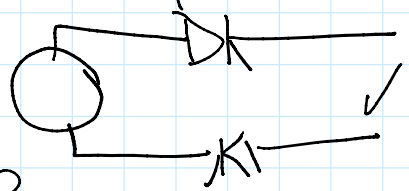
equivalent circuit



question:



ΔV_o :



Conclusion: to prevent the converters from entering a disc. mode $\rightarrow \Delta V_o$ must be biased ($+V_{dc1}, \Delta V_{dc}$)

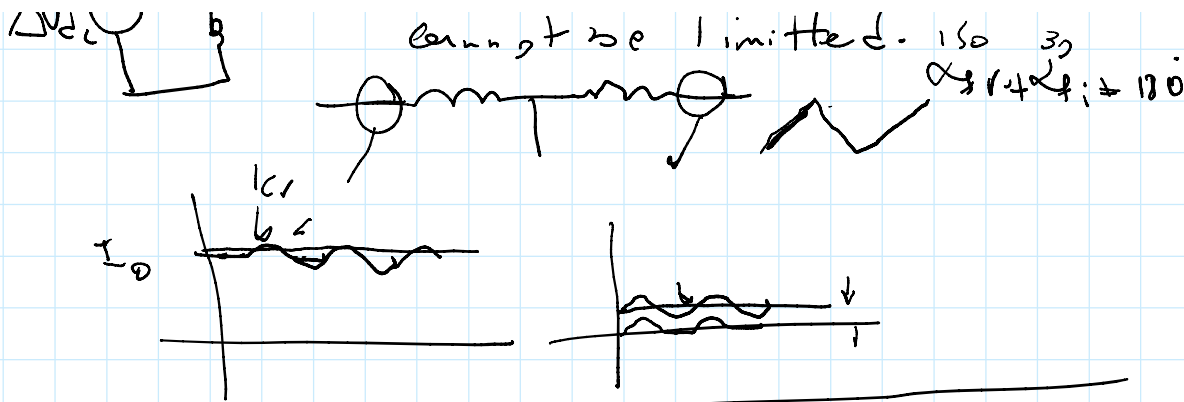
in other words $\Delta V_o = \bar{V}_{o1} + \bar{V}_{o2} \Rightarrow \alpha_1 + \alpha_2 < 180^\circ$

$L_1, L_2 \Rightarrow$ saturate $\rightarrow i_{cr}$

cannot be limited.

$\alpha_1 + \alpha_2 \neq 180^\circ$





example: DC-motor, 20hp, 300V, 900 rpm, separate d.c. supply

$$R_a = 0.25 \Omega, R_f = 145 \Omega, K_a = 1.2 \text{ V/A-rad/s}$$

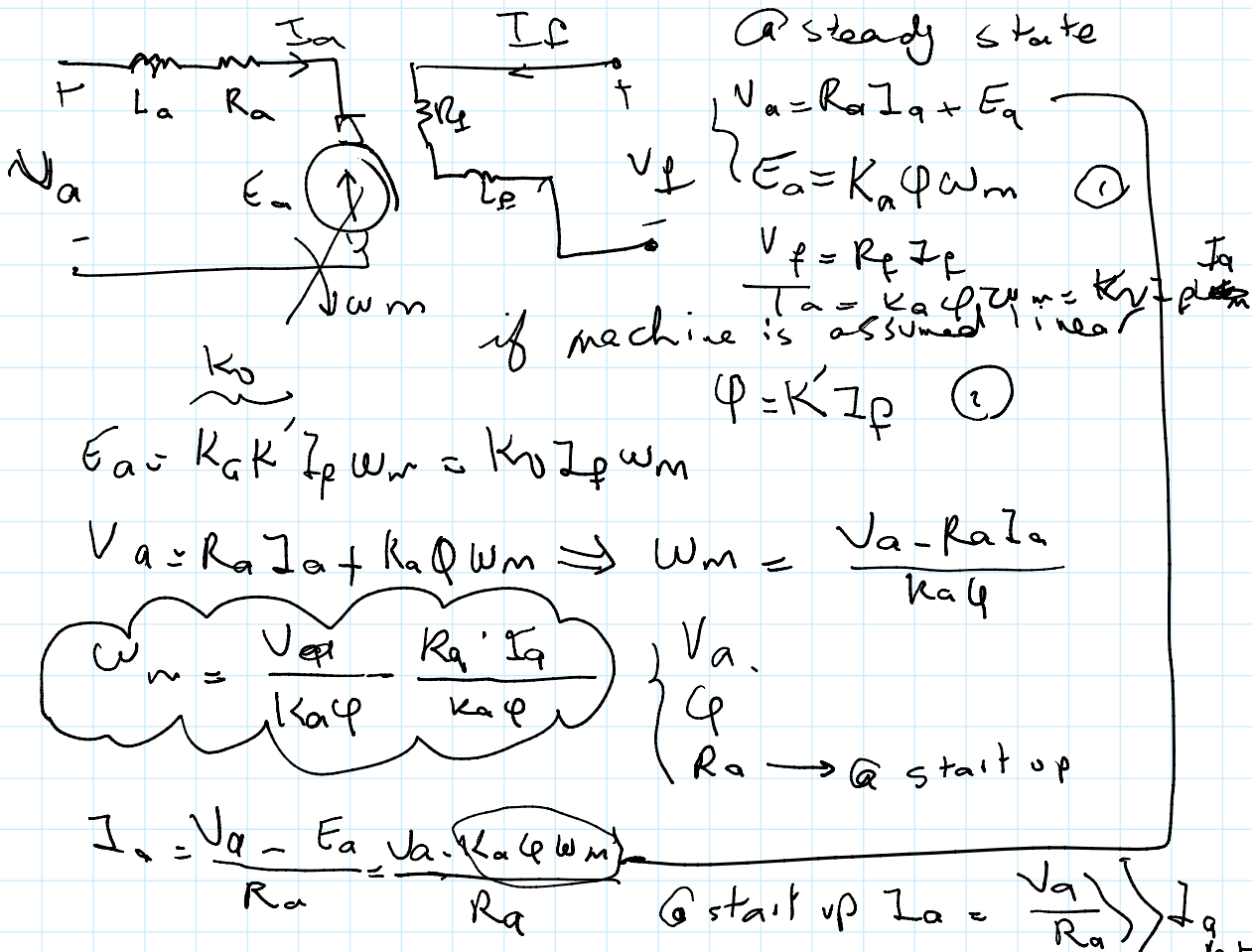
$$a.c.: 3\phi, 208 \text{ V}, 60 \text{ Hz}$$

mechanical losses are neglected

currents can be assumed ripple-free

a) if $T_L = 116 \text{ N.m}$, $\omega_m = 900 \text{ rpm}$
 determine the control strategy

Review on dc motor speed control:



$$\frac{V_a}{R_a} = \frac{V_a}{R_a} \quad \text{at start up } I_a = \frac{V_a}{R_a} \gg I_{a \text{ rated}}$$

$$\left\{ \begin{array}{l} \omega_m = 0 \text{ to } \omega_m = \omega_{\text{base}} \Rightarrow \boxed{V_a \text{ Control}} \\ I_a = I_{a \text{ max}}, I_f = I_{f \text{ max}} \end{array} \right.$$

normally a dc-motor is controlled at a constant rated armature current - $I_a = I_{a \text{ rated}}$ constant at rated

$$T_a = K_a \phi I_a = K_v I_f I_a = T_{a \text{ max}} = \text{constant} \\ \text{constant torque.}$$

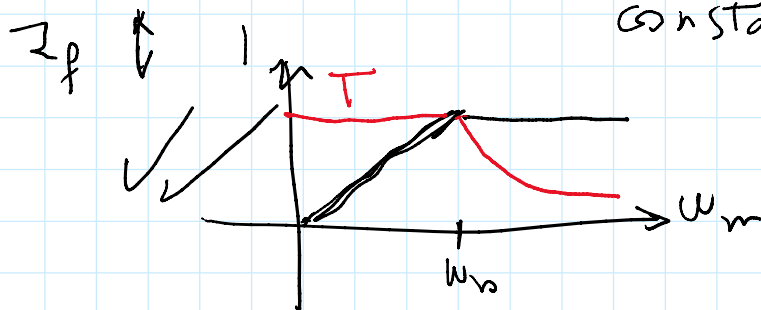
$$\left\{ \begin{array}{l} V_a = V_{a \text{ max}} \\ T_a = T_{a \text{ max}} = \text{constant} \Rightarrow \omega_m = \omega_{\text{base}} \end{array} \right. \left\{ \begin{array}{l} V_a = V_{a \text{ max}} \\ V_f = V_{f \text{ max}} \end{array} \right.$$

$$\omega_m = \frac{V_a}{K_a \phi} - \frac{R_a I_a}{K_a \phi} \approx \frac{V_a}{K_a \phi}$$

$$\omega_m > \omega_{\text{base}}$$

$\phi \downarrow, I_f \downarrow \rightarrow$ field weakeni

$$\left. \begin{array}{l} V_a = V_{a \text{ max}} \\ I_a = I_{a \text{ rated}} = \text{constant} \end{array} \right\} \rightarrow P_a = V_a I_a = \text{constant} \\ \text{constant power}$$

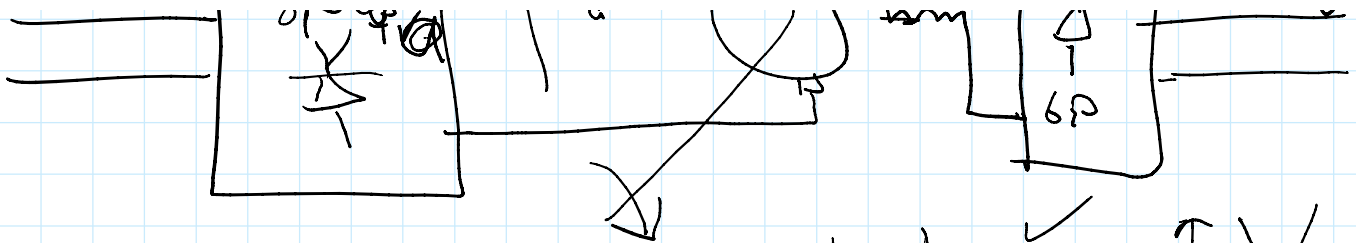


$$P = V_a I_a = \text{constant}$$

$$T = P / \omega_m$$

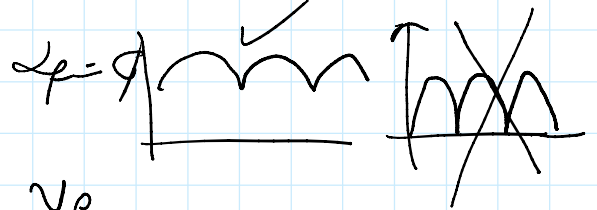
$$1) T_c = 116 \text{ Nm. } \quad \omega_m = 900 \text{ rpm}$$





$$T_a = K_a \phi \omega_m = K_v I_f \omega_m$$

$$= \frac{K_v}{R_f} V_f \omega_m$$



① draw the circuit

② determine ϕ_o , ϕ_f

Since $\omega_m = \omega_{rated} \ll \omega_{base} \Rightarrow$ - Constant torque
- Voltage

$$T_a = T_{aman}$$

$$V_f = V_{fmax}$$

$$\alpha_{ff} = \phi$$

$$V_f = \frac{3}{\pi} V_{up} \cos \phi = \frac{3}{\pi} V_{up}$$

$$\alpha_{fn} \Rightarrow V_a = \frac{3}{\pi} V_{up} \cos \phi_a = \frac{3}{\pi} 208\sqrt{2} \times 1 = 280.7 \text{ V}$$

$$V_a - R_a I_a + E = R_f I_a + K_v I_f \omega_m$$

$$I_f = \frac{V_f}{R_f} = \frac{280.7}{1.45} = 1.936 \text{ A}$$

$$T_a = K_a \phi I_a = K_v I_f I_a \Rightarrow I_a = \frac{116}{1.2 \times 1.936} = 49.93 \text{ A}$$

$$E_a = K_a \phi \omega_m = K_v I_f \omega_m = 1.2 \times 1.936 \times \frac{900}{180} \times 2\pi = 218.96 \text{ V}$$

$$V_a = 0.25 \times 49.93 + 218.96 = 231.44 \text{ V}$$

$$V_a = \frac{3}{\pi} V_{up} \cos \phi_a \Rightarrow \phi_a = 34.46^\circ$$

b) ω_{base} $V_a = V_{aman}$

$$b) \omega_{base} \left| \begin{array}{l} V_a = V_{amax} \\ I_f = I_{fmax} (V_f = V_{fmax}) \end{array} \right.$$

$$V_a = V_{amax} = \frac{3}{\pi} V_{up} \cos \phi_a = \frac{3}{\pi} V_{up} = 280.7 \text{ V}$$

$$V_a = E_a + R_e I_a = K_v I_f \omega_m + R_a I_a$$

$$\omega_m = \frac{V_{amax} - R_e I_a}{K_v I_{fmax}} = 115.36 \text{ rad/s} \quad \underline{1101.6 \text{ rpm}}$$

c) if $\omega_m = 1800 \text{ rpm}$, $\phi_a \neq \phi_f = 90^\circ$ } $\phi_a = \phi$
constant power — field weakening

$$\omega_m = 188.5 \text{ rad/s}, \quad E_a = V_a - R_e I_a = 280.7 - 0.25 \times 49.93$$

$$V_a = \frac{3}{\pi} V_{up} = 280.7$$

$$E_a = K_v I_f \omega_m \Rightarrow I_f = \frac{268.22}{1.2 \times 188.5} = 1.186 \text{ A}$$

$$V_f = R_f I_f = 121.5 \times 1.186 = 171.97 \text{ V}$$

$$V_f = \frac{3}{\pi} V_{up} \cos \phi_f \Rightarrow \boxed{\phi_{ff} = 52.2^\circ}$$

AC — AC converters

V_f $V \uparrow$ $f \uparrow$ $V \uparrow f \uparrow$ $\nwarrow$ direct AC-AC (one-converter)
 $\nearrow$ indirect AC-DC-AC (two-converter)

AC-AC $\begin{cases} f = \text{constant}, V \uparrow \downarrow = \text{AC Voltage controlled} \begin{cases} 1\phi \\ 3\phi \end{cases} \\ f \neq \text{constant} \text{ cycloconverters } \begin{cases} 1\phi \end{cases} \end{cases}$

$f \neq \text{constant}$ Cycloconverters $\begin{cases} 1\phi \\ 3\phi \end{cases}$